# The Link Function Problem in Psychological Interaction Testing

Enrico Toffalini[1], Margherita Calderan[1], Laura Sità[1], Tommaso Feraco[1], and Filippo Gambarota[2]

[1]Department of General Psychology, University of Padova

[2]Department of Developmental Psychology and Socialization, University of Padova

## Author Note

Enrico Toffalini https://orcid.org/0000-0002-1404-5133

Margherita Calderan https://orcid.org/0009-0005-5668-5162

Laura Sità https://orcid.org/0009-0000-6017-9979

Tommaso Feraco https://orcid.org/0000-0002-8920-5330

Filippo Gambarota https://orcid.org/0000-0002-6666-1747

Correspondence concerning this article should be addressed to Enrico Toffalini, Department of General Psychology, University of Padova, Via Venezia 8, Padova 35131, Italy, Email: enrico.toffalini@unipd.it

**Abstract**

Interaction tests are common in psychology, but an interaction claim is not defined by a product term alone. It also depends on the scale on which additivity is assumed. We reframe this scale problem in the GLM language of outcome family and link function: the family describes the response, whereas the link defines the scale on which no product-term interaction is assumed. We first report a preregistered descriptive review of recent psychological articles, showing that interaction tests are common and that family, link, and scale-of-additivity choices are often implicit when outcomes make identity-link additivity non-obvious. We then develop a link-centered framework that distinguishes wrong-family, wrong-link, and measurement-metric problems. The main focus is the correct-family, wrong-link problem: even when the response family is broadly plausible, different links can define different no-interaction baselines. Worked examples and simulations focus on forced-choice accuracy with a non-zero chance floor, within-family link choices such as logit versus probit, and sum scores as a related measurement case. The simulations show that plausible but inadequate links can turn additive main-effect structures on one scale into statistically significant pseudo-interactions on another. A diagnostic simulation shows that residual checks, link checks, and AIC comparisons may help, but they do not by themselves decide which scale is right for the scientific interaction claim. We argue that interaction claims are often underspecified unless researchers state, justify, and probe the scale on which no interaction is assumed.

*Keywords:* interactions, moderation, generalized linear models, link functions, model misspecification

## The Link Function Problem in Psychological Interaction Testing

Psychological theories often test whether the effect of one variable depends on another. Stress may impair performance more strongly among people with high anxiety. A treatment may help some participants more than others. A developmental difference may be large at one age and small at another. These are interaction claims: claims about moderation, boundary conditions, and heterogeneous effects. In practice, testing such claims has become almost a routine step after main effects have been examined. This routine use is understandable, because interactions often express important theoretical ideas. Yet an interaction claim should not be interpreted automatically from the presence or absence of a product term (Rimpler et al., 2026; Rohrer & Arslan, 2021).

Statistically, many interaction claims can be understood as claims about a difference between differences. Does the treatment-control difference change across groups? Does the age-related difference differ between populations? Does the effect of task difficulty vary as anxiety increases? While this formulation sounds intuitive, it hides a crucial assumption. An interaction is a difference between differences, but whether two differences are equal depends on the scale on which those differences are expressed.

The scale problem has been recognized for a long time. Loftus argued that interactions involving response probabilities are meaningful only relative to an assumed mapping between the observed response and the theoretical component of interest (Loftus, 1978). Cox treated interaction as a statistical concept tied to model formulation, transformation, and interpretation (Cox, 1984). Wagenmakers and colleagues later emphasized the distinction between removable and nonremovable interactions, showing that some interactions depend on the measurement scale and therefore do not support unambiguous conclusions about latent psychological processes (Wagenmakers et al., 2012). Busemeyer and Jones (1983) showed how conclusions about multiplicative combination rules depend on the measurement model and on measurement error in the observed variables. Lubinski and Humphreys illustrated how apparently substantive moderator effects can arise from model specification choices rather than real psychological synergy (Lubinski & Humphreys, 1990). More recently, Rohrer and Arslan argued that interaction tests can provide precise answers to vague questions when researchers do not specify the estimand, scale, or substantive meaning of the interaction [(**?**)

In psychology, this scale problem often has a measurement origin. Researchers usually reason about constructs such as ability, symptom severity, attitude strength, processing efficiency, or others as if they were continuous underlying dimensions. Most psychological dimensions may indeed be treated as smooth and roughly Gaussian, because they plausibly reflect many small sources of variation. If such processes were observed directly on an interval scale, a Gaussian identity model might often be a reasonable solution. But psychological phenomena are rarely observed directly. They are observed indirectly through tasks, items, ratings, counts, accuracies, and summed responses. These observations impose bounds, thresholds, discreteness, floors, ceilings, and chance levels. The link-function problem is one formal expression of this broader psychometric problem: the model must say how an additive structure on one scale is mapped onto the scale on which the data are observed.

A related literature has sharpened the problem for modern psychological data. Work on multiplicative interaction models has warned that interaction estimates can depend strongly on functional-form assumptions and on the observed range of the data (Hainmueller et al., 2019). Recent guidance on power and stability also shows that interaction estimates are often fragile for ordinary statistical reasons, such as small effects, low reliability, and limited power (Baranger et al., 2023; Castillo et al., 2026). Domingue and colleagues showed that misspecifying the outcome model can produce bias and false discoveries in interaction analyses, especially when outcomes are binary, count, or censored (Domingue et al., 2024). McCabe and colleagues showed that, in generalized linear models (GLMs) for probabilities and counts, product-term coefficients are not generally equal to interaction effects on the response scale (McCabe et al., 2022). Work on nonlinear probability models reaches the same practical conclusion: interaction effects should often be evaluated as conditional marginal effects or response-scale contrasts, not as product coefficients alone (Ai & Norton, 2003; Mize, 2019). More broadly, estimand-centered approaches recommend treating models as tools for generating substantively meaningful quantities rather than interpreting coefficients mechanically (Rohrer & Arel-Bundock, 2026). Rimpler and colleagues further emphasized that a theoretical claim of conditionality does not automatically imply that the correct statistical representation is a multiplicative product term (Rimpler et al., 2026). Together, these contributions show that interaction testing is not only a matter of adding the right-hand-side product term to a regression equation.

The present paper focuses on one specific part of this broader problem: the role of the link function in GLM interaction testing. We do not claim that scale dependence of interactions is new (Cox, 1984; Loftus, 1978; Wagenmakers et al., 2012), or that product terms in GLMs can be read directly as response-scale interactions (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019). Our contribution is narrower. We translate the scale-dependence problem into the GLM distinction between outcome family and link function, and focus on how link choice can change the no-interaction baseline even when the analyst has already moved beyond a simple Gaussian identity model. The core case is correct family, wrong link: a model may respect the outcome support and sampling process, but still define an inadequate scale of additivity for the interaction claim. The review documents how often these choices are implicit in recent psychological interaction tests. The worked examples and simulations then show how plausible link choices can generate pseudo-interactions under controlled conditions. Related wrong-family and measurement-metric problems are kept visible because they often coexist in psychological data, but they are not the main focus of th

Choosing an appropriate family is therefore necessary, but not sufficient, for appropriate interaction testing. Two models can respect the same outcome family while defining different no-interaction baselines. For response times, for example, a Gamma family may be plausible because the outcome is positive, continuous, and right-skewed. Yet `Gamma(link = "log")` assumes additivity in log mean response times, which corresponds to proportional change on the millisecond scale, whereas `Gamma(link = "inverse")`, the canonical link, assumes additivity in inverse mean response times. Neither choice is automatically correct. The relevant question is whether the scientific claim concerns proportional change, reciprocal change, absolute change, or some other scale. Link misspecification can affect likelihood-based inference in GLMs and GLMMs (Czado & Santner, 1992; Yu & Huang, 2019).

The same logic applies to binary and accuracy data, which are frequently reported in psychological research. A binomial family may correctly represent the sampling process of correct and incorrect responses, but the link still defines the scale of additivity. A standard logit link maps the linear predictor to the full 0-to-1 probability range, and a probit link makes a similar but not identical mapping, with different curvature especially away from the middle of the probability scale. In a two-alternative forced-choice task, however, the task-implied lower bound for performance above guessing is often 0.50. A chance-corrected link can encode this lower asymptote directly when that is the scale required by the scientific claim. Thus, the distinction is not only between linear models and GLMs. It is also across plausible links within a generally appropriate family.

Figure 1 shows why link choice is a scale choice, not only a curve-fitting choice. In each panel, the grey construction lines are anchored to a target link: the chance-corrected logit for forced-choice accuracy and the log link for Gamma mean response times. Equal steps on the target-link scale map to unequal steps on any other scale. Conversely, the same observed response values would not be equally spaced on an alternative link scale. Floor and ceiling effects are the most visible version of this problem, because the response scale has almost no room left to move. But the same logic is not limited to the endpoints: whenever the mapping from the linear predictor to the response is nonlinear, the amount of response-scale change implied by the same link-scale effect varies across the scale. This is the central reason why link choice matters for interactions. A data-generating process without a product term is matched by an additive model only on the scale defined by the correct link. When another plausible link is used, the same response-scale pattern can imply a different no-interaction baseline and therefore a different moderation claim.

Figure 2 uses a simulated two-alternative forced-choice accuracy dataset as an illustrative example. Each participant completes 50 trials, and the outcome is the proportion correct. The predictors are age and group. The data-generating model includes an age effect and a group effect, but no age-by-group term on the chance-corrected logit scale, the scale on which the task has a 0.50 chance floor. The figure illustrates how the same data can apparently support different moderation claims when fitted models define different no-interaction baselines (Cox, 1984; Domingue et al., 2024; Loftus, 1978; McCabe et al., 2022), that is, when they use different link functions.

This does not mean that all interactions are artifacts, or that scale choices make any interaction claim arbitrary. Some interactions are robust across any link function [e.g., non-removable or crossover interactions; Loftus (1978); Cox (1984); Wagenmakers et al. (2012)]. The problem is that many interaction claims are underspecified unless researchers state which scale defines additivity, whether the reported interaction is on the linear predictor scale or the observed response scale, and whether alternative plausible links lead to the same substantive conclusion. In this sense, link choice is not just a technical detail, but it becomes part of the scientific claim.

The paper proceeds in four steps. First, the preregistered review describes current reporting practice: how often interaction tests are used, how often family and link choices are explicit, and how often outcomes make identity-link additivity non-obvious. Second, the framework separates outcome family, link function, and measurement metric, with special attention to the correct-family, wrong-link case. Third, the applied sections develop three psychology-relevant cases: forced-choice accuracy with a non-zero chance floor, sum scores, and within-family link choice. Fourth, the simulations quantify when link or metric choices can generate pseudo-interactions, and the diagnostic example asks what standard checks can and cannot detect.

## Current Practice in Psychological Interaction Testing: A Preregistered Review

### Why This Review Matters

Do these issues matter for current psychological practice? This question motivated our preregistered descriptive review. The review examined whether the link-function problem is relevant to the kinds of interaction analyses that psychologists actually report. If interaction tests were rare, or if researchers usually stated the outcome family, link function, and scale of additivity, the problem would be less pressing. If, instead, interaction tests are common and these modeling choices are often implicit, then many moderation claims are difficult to evaluate, because the no-interaction baseline is not fully specified.

The present review is not an audit of individual articles. We did not aim to decide whether each reported interaction was right or wrong, or whether it was removable or nonremovable. Such judgments often require more information than published reports provide, such as detailed interaction plots, complete model coefficients, the assumed measurement scale, the link function, or the substantive scale on which the interaction was meant to hold. Although this distinction is central in the classical scale-dependence literature (Cox, 1984; Loftus, 1978; Wagenmakers et al., 2012), we did not systematically code it because it could not be reconstructed reliably from many published reports. Instead, we treated the review as a description of reported practice. We focused on what could be observed from the articles: how often interaction tests were used, whether link functions were stated explicitly, and how often the outcomes were of a kind for which identity-link additivity is not an obvious default. The broader implication is that many interaction claims are partly underspecified when the outcome family, link function, and scale of additivity are left implicit. Making these choices explicit is part of good quantitative reporting, because readers need enough information to understand which model was fitted and which kind of interaction was tested (Mustillo et al., 2018).

## Review Method

The review protocol was preregistered before the records were screened, at: xxxxxxxxxxxxxx. We specified a descriptive review of recently published psychological research articles, with four primary goals: estimating how often empirical articles test interactions, how often interaction-testing articles use non-identity links, how often the link function is explicitly reported, and which response-variable types are used in interaction analyses. The protocol also stated that the review was not intended to be an exhaustive systematic review of psychology as a whole, but a targeted estimate of current practice in selected journals.

Sampled articles were recent publications, all from 2024, in five prominent journals: *Psychological Science*, *Journal of Experimental Psychology: General*, *Journal of Personality and Social Psychology*, *Developmental Science*, and *Psychological Medicine*. These journals were pre-specified by the authors to cover different areas of psychology. Eligible articles were empirical articles presenting original psychological research. Reviews, meta-analyses, editorials, theoretical articles without empirical data, qualitative-only articles, replication studies, and psychometric validation studies were excluded. After preregistration, we also excluded articles in which the relevant tests were conducted only at the latent-variable level, or in which the empirical analyses relied only on structural equation models (SEM). This decision kept the review focused on observed outcomes analyzed with regression, ANOVA, or GLM-like models. SEM is relevant to the broader issue because it can make the measurement model explicit and may therefore help address some scale problems. At the same time, SEM raises additional questions about links, thresholds, loadings, and latent-response metrics that would require a different coding scheme and are beyond the scope of the present review. Articles were then coded in detail if they tested at least one interaction effect, either through statistical modeling or analysis of variance.

For each interaction-testing article, coders recorded whether at least one tested interaction used a non-identity link function, whether the link function was explicitly reported, whether at least one interaction involved a potentially questionable identity-link analysis, whether at least one such interaction was statistically significant or interpreted analogously, and which response-variable types were analyzed. We use the term potentially questionable identity-link analysis for cases in which an interaction was analyzed with an identity link although the reported outcome made observed-scale additivity non-obvious. The term is descriptive and cautious: it does not mean that the analysis was necessarily wrong, but that the scale of additivity was not self-evident from the published report. The protocol specified descriptive summaries, including proportions and Wilson 95% confidence intervals for Boolean variables. The full review protocol, coding definitions, and extended descriptive tables are reported in the Supplement.

**Review Findings**

The review identified 331 eligible empirical articles. Interaction testing was common: 200 of these articles tested at least one interaction, corresponding to 60.4%, 95% CI [55.1, 65.5]. Among the 200 interaction-testing articles, 58 used at least one non-identity link function, 29%, 95% CI [23.2, 35.6], and only 46 explicitly reported the link function, 23%, 95% CI [17.7, 29.3]. Thus, in most interaction-testing articles, the link function was either identity by default, not reported explicitly, or not foregrounded as part of the interaction claim.

A large proportion of interaction-testing articles involved what we call potentially questionable identity-link analyses. Among the 200 articles testing interactions, 144 included at least one identity-link analysis in which the outcome made observed-scale additivity non-obvious, 72%, 95% CI [65.4, 77.8]. This code should be interpreted cautiously. It does not show that the fitted model was wrong, or that the specific interaction claim was false. It shows that the scale of additivity was difficult to evaluate from the reported outcome and model description. This is the same broad concern raised by work on outcome-model mismatch and metric assumptions in interaction testing (Domingue et al., 2024; Liddell & Kruschke, 2018). Among these 144 cases, 124 reported at least one statistically significant interaction, 86.1%, 95% CI [79.5, 90.8]. This last number should not be read as a false-positive estimate. It shows how often potentially questionable identity-link analyses led to interaction claims that could matter for interpretation. Table 1 summarizes these descriptive results.

The outcomes used in interaction analyses were heterogeneous. The most common broad, non-exclusive categories were binary, accuracy, or proportion outcomes (86 articles), sum scores or composites (77 articles), ordinal, Likert, or rating outcomes (49 articles), response times or durations (29 articles), and counts or error counts (12 articles). This heterogeneity matters because the identity link has different implications across outcome types. For some approximately continuous outcomes, additivity on the observed scale may be a defensible target. For bounded, discrete, ordinal, chance-constrained, or heavily transformed outcomes, however, additivity on the observed scale is an assumption that should usually be stated and examined. This is especially clear for ordinal and Likert-type outcomes, where treating ordered categories as metric can affect estimates, error rates, and interaction conclusions (Bürkner & Vuorre, 2019; Liddell & Kruschke, 2018).

### What the Review Shows, and What It Does Not Show

The review motivates the framework rather than proving that specific analyses were wrong. It shows that interaction testing is a routine part of current psychological practice, that link functions are rarely made explicit, and that many interaction claims are made with outcomes for which the scale of additivity is not trivial. This combination is the key empirical result. If the link function defines the scale on which effects are assumed to add, then an implicit link function implies an implicit no-interaction baseline.

The review does not show that the coded articles reached wrong conclusions. Some identity-link analyses may have been reasonable approximations. Some interaction claims may be robust across plausible links. Some authors may have intentionally targeted observed-scale differences, even if this was not stated in link-function language. The review also does not classify individual interactions as removable or nonremovable. That classification would often require information that published reports do not provide, including the raw outcome scale, the fitted link or transformation, the shape of the interaction, and the substantive scale of interest. The descriptive result is more modest, but still important: many published interaction claims include potentially questionable identity-link analyses because the family, link, and scale of additivity are not made sufficiently explicit from the reader's perspective.

The next section turns this reporting issue into a statistical framework. We first distinguish the outcome family from the link function, then show why choosing the family is not enough for interaction testing, and finally separate interactions on the linear-predictor scale from interactions on the observed response scale.

## The Link Function Problem

The review described a reporting problem: interaction tests are often presented without an explicit statement of the outcome family, link function, or scale of additivity. We now formalize why this matters. The main issue is not simply that some outcomes are nonnormal, bounded, or discrete. It is that the model defines the scale on which effects are assumed to add. In GLMs, this scale is defined by the link function. This matters especially when psychological data are indirect observations of a target process. A construct may be conceived as continuous, but the observed response is produced by a measurement device, task, or scoring rule that can remap that construct onto a constrained scale. Interaction tests are therefore interpreted relative to the scale imposed by the fitted model.

In a generalized linear model (GLM), the expected outcome for observation $i$, denoted $\mu_i = E(Y_i)$, is related to a linear predictor through a link function (McCullagh & Nelder, 1989; Nelder & Wedderburn, 1972):

$$g(\mu_i) = \eta_i.$$

With two predictors, $X_i$ and $Z_i$, and their product term, the linear predictor is written as

$$\eta_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i + \beta_3 X_i Z_i.$$

The first equation contains the link function, $g(\cdot)$. This is the function that maps the conditional mean of the outcome, $\mu_i$, onto the linear predictor, $\eta_i$. The second equation then specifies what is additive on that linear-predictor scale: the intercept, the two main effects, and, if included, the product term.

This separation is important because a GLM involves two related but different choices. First, the analyst chooses a family, that is, a model for the kind of outcome being analyzed. This choice defines the possible values of the response and its sampling process, for example Gaussian scores, binomial successes, Poisson counts, or positive continuous outcomes. Second, the analyst chooses a link function, that is, the function $g(\cdot)$ that defines the scale on which the predictors are combined additively.

This second choice is crucial for interaction testing. In a model without the product term, $X_i$ and $Z_i$ are additive on the $\eta_i$ scale, not necessarily on the observed outcome scale. The link function defines how this additive structure is translated back to the response scale. For this reason, choosing a plausible family is necessary but not sufficient. The family can keep predictions within the right range, but the link still defines the scale on which the product term is interpreted as a departure from additivity.

**What the link function does**

The linear predictor, $\eta$, is the model's additive scale. Intercepts, main effects, product terms, covariates, and random effects are combined before the inverse link is applied. Many psychological outcomes, however, are not direct readings of the psychological continuum of interest. They are observed through constrained measurement scales. Counts cannot be negative. Probabilities must lie between 0 and 1. Forced-choice accuracy may have a theoretical floor above 0, for example .50 in a true/false task. Response times are positive and often right-skewed. Sum scores have lower and upper bounds, and are often discrete.

The inverse link, $g^{-1}(\eta)$, maps the additive model back to the expected response. This mapping is not neutral for interactions. Equal changes in $\eta$ most often become unequal changes in probabilities, counts, milliseconds, or score units. Conversely, equal observed-scale differences need not be equal differences on the link scale. The link function therefore defines what "no interaction" means in the fitted model.

The distinction is hidden in ordinary linear models because the identity link makes the two scales coincide. If $g(\mu) = \mu$, then additivity on the linear predictor scale is also additivity on the response scale. For any other link function, this equivalence disappears. A logit link defines additivity in log odds. A probit link defines additivity on a latent-normal scale. A log link defines additivity in log means (which corresponds to multiplicative change on the response scale). The same general point applies within a family. As noted in the introduction, a Gamma model for response times may use an identity, log, or inverse (the canonical) link: the family is the same, but the implied no-interaction baseline is different across link choices.

A simple count example

Suppose children make fewer errors as they grow up. A Poisson model with a log link can represent this as a linear age effect on the log mean,

$$\log\{E(Y)\} = \beta_0 + \beta_1 \text{age},$$

with $\beta_1 < 0$. On the observed count scale, the same model implies $E(Y) = \exp(\beta_0 + \beta_1 \text{age})$, a curved decreasing function. The curvature is not an extra psychological mechanism, but the expected consequence of mapping an additive model on the log scale back to non-negative counts.

Now add a group/condition difference in overall error rate, but no age-by-condition product term on the log scale. On the observed count scale, the absolute age-related reduction will usually be larger where expected errors are high and smaller where expected errors are low. A model that assumes additivity on another scale, especially an identity link, can treat this changing absolute difference as moderation. The same logic applies within the same family: a Poisson model with an identity link and a Poisson model with a log link both model counts, but they define different no-interaction baselines. The point is not that Poisson-log is always correct. The point is that a product term is a departure from additivity on the fitted link scale, whereas response-scale differences may change even when there is no product term on the generating scale (Domingue et al., 2024; McCabe et al., 2022).

The same logic applies directly to interaction coefficients. The coefficient $\beta_3$ is a product-term interaction on the link scale, $g(\mu)$. When $\beta_3 = 0$, the fitted model assumes additivity on that scale. It does not necessarily assume additivity on the observed response scale. A response-scale interaction can instead be defined as a difference between expected differences. For two binary predictors, one such quantity is

$$\{E(Y \mid X = 1, Z = 1) - E(Y \mid X = 0, Z = 1)\} - \{E(Y \mid X = 1, Z = 0) - E(Y \mid X = 0, Z = 0)\}.$$

In a nonlinear GLM, this quantity depends on the inverse link, $g^{-1}(\eta)$. It can be nonzero when the product-term coefficient is zero, and it can differ in size or sign from the product-term coefficient when the product term is nonzero. This is why product-term coefficients in GLMs should not be treated as automatic answers to response-scale moderation questions (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019).

The link-function problem is therefore a scale-of-additivity problem. An interaction claim is always relative to a scale: it says that the effect of $X$ is not constant across levels of $Z$ when effects are expressed on that scale. An identity-link model evaluates this claim on the expected observed mean. A binomial-logit model evaluates it in log odds. A binomial-probit model evaluates it on a probit scale. A chance-corrected binomial link evaluates it on a scale that maps the linear predictor onto the interval from chance performance to perfect performance. These are related statistical models, but they do not express the same scientific null.

This point helps separate two modeling problems that are often conflated. A wrong-family problem occurs when the model does not respect the outcome type, for example when counts, proportions, or trial accuracies are treated as unbounded Gaussian observations without justification. A wrong-link problem occurs when the broad family is plausible, but the link defines an inadequate scale of additivity, for example when a standard binomial logit link is used for a forced-choice task with a non-zero chance floor. Some outcomes add a further measurement complication: the recorded score may not be an interval-scale measure of the target construct, as with ordinal responses or some sum scores. These issues can overlap, but the focus of this paper is the link function, especially cases in which the family is broadly plausible but the scale of additivity remains underjustified.

Our paper focuses mainly on the second problem, while keeping the other two visible. This focus is narrower than the general warning that linear models can be inappropriate for non-Gaussian outcomes. Researchers may choose a broadly appropriate family and still leave the scientific scale of additivity unspecified. For instance, a binomial family may correctly represent successes out of trials, but it does not choose among logit, probit, complementary log-log, flexible generalized links, or a chance-corrected link with a lower asymptote (Aranda-Ordaz, 1981; Stukel, 1988). Link misspecification can affect inference in binary GLMs and GLMMs, and diagnostic evidence may not identify a unique correct link (Czado & Santner, 1992; Yu & Huang, 2019).

A link-aware interaction analysis therefore begins with four questions. First, what is the estimand: a link-scale coefficient, a probability difference, a marginal effect, a risk ratio, a count difference, or another response-scale contrast? Second, what constraints does the outcome impose, such as bounds, discreteness, ordinal categories, skew, ceilings, floors, or chance performance? Third, which scale of additivity is scientifically plausible for the process under study? Fourth, does the substantive conclusion survive plausible alternative links? These questions do not make interaction testing automatic, but they make the assumptions visible enough to evaluate.

This framework also clarifies the role of the simulations that follow. The simulations do not ask whether every interaction in psychology is an artifact. They ask whether an interaction can be generated, removed, or distorted when the fitted model assumes additivity on a different scale from the data-generating process. This matters because many psychological outcomes have exactly the properties that make link assumptions consequential: bounded accuracy, non-zero chance levels, ordinal ratings, counts, response times, and sum scores. The next sections develop these cases in turn.

**Box 1. Prediction, Explanation, and the Scale of Interaction**

The importance of link choice depends on the goal of the analysis. For purely predictive purposes, the primary criteria may be predictive accuracy, calibration, generalizability, and decision usefulness. A model can be useful if it predicts future observations well, even if its coefficients do not provide a faithful description of the psychological process that generated the data.

Scientific interaction claims require a different standard. When researchers claim that an effect is moderated, they usually claim more than that one model predicts slightly better than another. They claim that the effect of one variable changes across levels of another variable in a way that is meaningful for theory. That claim requires an estimand. It also requires a scale. A treatment effect might be larger in absolute probability points, larger in odds ratios, larger on a latent response scale, or larger only after a transformation. These are different claims, and they can lead to different conclusions.

This distinction is consistent with estimand-centered workflows that treat models as tools for generating clear target quantities rather than as collections of coefficients to be interpreted mechanically (Rohrer & Arel-Bundock, 2026). For interaction testing, the relevant target quantity should be stated before the product term is interpreted. If the target is a response-scale difference-in-differences, the analyst should compute and report that quantity. If the target is additivity on a latent or link scale, the link function should be justified as part of the scientific model.

The point is not that prediction and explanation are separate in practice. Predictive checks, calibration, and out-of-sample performance can all inform scientific modeling. The point is that good prediction alone does not specify the scale on which a moderation claim is being made. For scientific inference about interactions, the link function is part of the claim.

## Forced-Choice Accuracy and the Non-Zero Chance Floor

Forced-choice accuracy is a useful case because it separates the outcome family from the link scale. If a participant completes $k_i$ trials and answers $Y_i$ trials correctly, the binomial sampling model is often the natural starting point:

$$Y_i \sim \text{Binomial}(k_i, p_i),$$

where $p_i$ is the expected probability of a correct response. This choice concerns the sampling process. It does not, by itself, decide the lower bound that is meaningful for the psychological task. In a 2-alternative forced-choice task, or in a true/false task when random responding between two alternatives is the relevant chance process, guessing implies an expected accuracy of .50. More generally, in an $m$-alternative forced-choice task, the chance level is $c = 1/m$. For these tasks, the expected performance curve may have a lower asymptote at chance, rather than at 0, when the target claim concerns performance above guessing.

This distinction gives three increasingly specific specifications. The weakest is a Gaussian identity model fitted to proportions. It ignores the binomial trial process and treats accuracy as an approximately continuous outcome. It also assumes additivity directly in percentage-correct units and can generate fitted values outside the admissible 0-to-1 range. This model may sometimes be useful as a rough descriptive approximation, especially far from the boundaries, but it does not encode either the trial structure or the task-implied lower bound.

A standard binomial logit or probit model improves on this by respecting the binomial sampling process and constraining fitted probabilities to the interval from 0 to 1:

$$p_i = F(\eta_i),$$

where $F$ is the logistic cumulative distribution function for a logit link, or the standard normal cumulative distribution function for a probit link. This model is right about the binomial structure of the data. However, in a forced-choice task with a known chance level, it still defines the lower asymptote as 0. A standard binomial model can respect the sampling process while encoding a lower asymptote that is not aligned with the task-based interpretation of performance above chance.

A chance-corrected binomial link keeps the binomial sampling model but maps the linear predictor to the interval from chance to 1:

$$p_i = c + (1 - c)F(\eta_i).$$

For a 2-AFC or true/false task with chance level .50, this becomes

$$p_i = .50 + .50F(\eta_i).$$

Equivalently, the link is applied to the above-chance quantity $(p_i - c)/(1 - c)$ rather than to the raw probability $p_i$. The no-interaction baseline is therefore different. A product term in a standard binomial-logit or binomial-probit model tests additivity on a scale defined over the full 0-to-1 probability range. A product term in a chance-corrected binomial model tests additivity on a latent performance-above-chance scale that already encodes the task-implied lower asymptote. These are different modeling claims, even though both use a binomial family.

The difference is easiest to see when one group, condition, or region of the predictor space lies close to chance, but it is not only a problem of obvious floor effects. Suppose that age has the same effect in two groups on the latent performance-above-chance scale, with no true age-by-group product term on that scale. On the observed accuracy scale, the lower-performing group can still show stronger compression near .50, simply because there is less meaningful room below chance. A standard logit or probit link places this compression near 0 rather than near .50. The fitted model can then use an age-by-group product term to absorb systematic curvature created by the misplaced lower bound. More generally, the inverse link changes the amount of response-scale room available for an effect across the scale, not only at the endpoints. The resulting interaction is therefore not merely random Type I error. It is a pseudo-interaction produced by a mismatch between the scale used by the model and the scale implied by the task. This is consistent with broader work showing that outcome-model misspecification can create false discoveries in interaction analyses (Domingue et al., 2024), with work showing that product-term coefficients in nonlinear probability models are not automatically response-scale interaction effects (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019), and with evidence that binary-link misspecification can affect likelihood-based inference (Czado & Santner, 1992; Yu & Huang, 2019).

This does not mean that every standard binomial model for accuracy is wrong. The chance-corrected link is most defensible when the chance level is known from the task design, guessing is a plausible lower-bound process, and below-chance performance is not substantively meaningful for the target claim. In the simplest case, the chance level can be fixed by design, for example $c = 0.50$ in a 2-AFC task. In other cases, analysts may estimate the lower asymptote or report sensitivity analyses over plausible values of $c$, although estimating it requires enough information and stronger assumptions. Richer models may also be needed if there are lapses, response biases, item heterogeneity, learning across trials, systematic below-chance performance, or participant-specific guessing strategies. The point is not to replace every standard binomial model with a chance-corrected model. The point is to ask whether the link used for the interaction test encodes the scale on which no interaction is scientifically meaningful.

The chance-corrected GLM link is closely related to psychometric-function models, where lower asymptotes, guessing, lapses, and goodness-of-fit checks are part of the response model, and to item response models that include a lower asymptote (Knoblauch & Maloney, 2012; Wichmann & Hill, 2001). For example, the 3-parameter logistic IRT model includes a guessing parameter at the item level, rather than treating the chance floor as a single aggregate constraint. When item-level or trial-level data are available, and when the target claim concerns latent performance rather than an aggregated accuracy proportion, such measurement models or trial-level GLMMs may be more appropriate than a single aggregate GLM (Moscatelli et al., 2012). The chance-corrected link used here should therefore be understood as a GLM-level representation of a design constraint. It is not a replacement for item-level measurement models when the data and the scientific question call for them.

The same dataset can also be analyzed for different purposes. If the goal is prediction, standard and chance-corrected binomial models can be compared by predictive accuracy, calibration, and generalizability. If the substantive estimand is an observed-scale quantity, such as a difference in percentage correct or a difference between such differences, that quantity should be reported directly as a response-scale contrast. If the claim concerns latent performance above chance, ability, sensitivity, or propensity to answer correctly beyond guessing, a chance-corrected link is often closer to the scientific claim. Thus, for interaction claims, the key question is not only whether accuracy is binomial. It is whether the model defines additivity on the scale required by the task and by the claim (Loftus, 1978; McCabe et al., 2022; Wichmann & Hill, 2001).

## Sum Scores as Bounded and Discrete Outcomes

Sum scores are not always a wrong-link problem in the narrow GLM sense. They are often a measurement-metric problem. We include them because metric assumptions and link assumptions create the same risk for interaction claims: they define the scale on which additivity is tested.

Sum scores make the link-function problem inseparable from measurement. Many psychological outcomes are reported as totals or averages across questionnaire items, task items, symptoms, errors, or ratings. Recent debates on sum scores make the same point from a psychometric angle: a total score can be useful and transparent, but it also encodes assumptions about how item responses represent the target construct (McNeish, 2023; McNeish & Wolf, 2020; Widaman & Revelle, 2023). These scores are often treated as if they were direct observations of an approximately continuous psychological dimension. But the latent construct and the observed score are different objects. The construct may be conceived as a continuous dimension, and may sometimes be reasonably approximated as smooth or roughly normal. The sum score is instead produced by a finite set of item responses through thresholds, endorsement probabilities, success probabilities, category labels, and scoring rules. It is therefore a remapped version of the target continuum, not the continuum itself.

This point matters even when the construct is otherwise well behaved. A normally distributed latent variable can still generate a bounded, discrete, skewed, or non-interval observed score. A total based on binary or ordinal items has only a limited number of possible values. It has a lower and upper bound. It can be skewed when items are too easy, too hard, rarely endorsed, or almost always endorsed. It can be compressed near floor or ceiling, but compression is not restricted to the endpoints. Adjacent raw-score values may correspond to different distances on the latent scale at different regions of the score range. These features are common in psychological variables (Micceri, 1989), and they are central to why treating ordinal responses as metric can produce false alarms, missed effects, reversals, and misleading interactions (Liddell & Kruschke, 2018).

This does not mean that sum scores are always wrong or useless. Sum scores can be descriptively useful, transparent, reliable enough for many purposes, comparable across studies, and pragmatically tied to how instruments are scored in applied settings (Widaman & Revelle, 2023). The narrower point is that a Gaussian identity model for a sum score is not neutral. It assumes that effects are additive in raw-score units on the observed score scale. In an interaction model, the no-interaction baseline is that the expected raw-score effect of one predictor is constant across levels of the other predictor. Sometimes this is the target estimand. For example, a clinician, teacher, or researcher may care about a two-point difference on the reported score because decisions and interpretations are made on that metric.

Gaussian identity models are more defensible when the score has many possible values, the distribution is not strongly skewed, observations are not concentrated near floor or ceiling, residual patterns are not strongly tied to fitted values, and the substantive claim is explicitly about raw-score differences. They are less defensible when the score has few levels, is built from ordinal item responses with unclear spacing, shows floor or ceiling concentration, or is interpreted as evidence about moderation on a latent construct scale.

The problem becomes sharper when the verbal claim moves from the manifest score to the underlying construct. A group near the upper bound may show a smaller observed treatment effect not because the treatment is less effective on the construct, but because the score has less room to increase. A group located near a dense region of thresholds may show a larger raw-score change because the instrument is more sensitive there. The product term can then partly reflect the measurement metric rather than genuine moderation in the psychological process. A closely related problem has been shown in item-level work on genotype-by-environment interactions, where summed item scores can mask genuine interactions or generate spurious ones through scale properties of the observed score (Molenaar & Dolan, 2014). This is the sum-score version of outcome-model and measurement-metric misspecification (Domingue et al., 2024).

Several pragmatic alternatives can make the scale assumption more explicit. Cumulative-link models treat the score as ordered categories and avoid assuming equal raw-score distances. Item-level ordinal or IRT-inspired models separate the latent construct from thresholds, item difficulties, and item discriminations. Bounded-count approximations, such as binomial, beta-binomial, or Poisson-binomial models, can be useful when the score is interpretable as a count of successes, symptoms, or endorsements. None of these alternatives is automatically correct. Each makes different assumptions, and each can fail. The point is also consistent with the sum-score debate: replacing a total score with a latent-score or item-level model changes the assumptions, rather than removing them (McNeish, 2023; McNeish & Wolf, 2020; Widaman & Revelle, 2023). The role of our proposed alternatives here is not to establish the true scale, but to test whether the interaction conclusion depends on the manifest sum-score metric (Bürkner & Vuorre, 2019; Domingue et al., 2024; Liddell & Kruschke, 2018).

The recommendation is therefore not to abandon sum scores. It is to make the assumed scale visible. Researchers can inspect the score distribution, plot predicted values near the bounds, report response-scale contrasts, and compare Gaussian identity results with ordinal, bounded-count, or item-level alternatives when these are plausible. If the interaction is stable across these specifications, the claim is easier to defend. If it appears only on the manifest sum-score metric, the conclusion can still be informative, but it should be framed as a property of that score, not automatically as moderation of the underlying psychological construct.

## Within-Family Link Choice

The link-function problem also arises within the same outcome family. For binary, binomial, ordinal, count, or positive continuous outcomes, the family can be broadly appropriate while the link remains underjustified. This is easy to miss because many default links are reasonable in many datasets. Logit and probit links, for example, often produce very similar fitted probabilities in the middle of the response range, especially after the usual approximate rescaling of coefficients. That practical similarity is real. It explains why many applied conclusions are unchanged by switching between the two links. However, it is not a general guarantee, especially when the claim depends on interaction, curvature, floor effects, ceiling effects, or tail behavior. Link misspecification can affect likelihood-based inference in binary GLMs and GLMMs (Czado & Santner, 1992; Yu & Huang, 2019).

The choice between logit and probit is therefore not only a matter of coefficient rescaling. A logit link is natural when the substantive interpretation is in odds or odds ratios. In that case, additivity on the linear predictor scale means additivity in log odds, and exponentiated coefficients have a direct odds-ratio interpretation. A probit link is natural when the response can be represented as threshold crossing on an approximately normal latent variable, such as ability, signal strength, response tendency, propensity, or latent correctness. This does not make probit generally superior to logit. It means that the two links encode different stories about the scale on which effects combine.

For interaction claims, this distinction is more consequential than it may appear from main effects alone. Two links can give nearly overlapping fitted probabilities in the central range but imply different distances on the link scale near the floor, ceiling, chance level, or distributional tails. A product term in a logit model asks whether effects depart from additivity in log odds. A product term in a probit model asks whether they depart from additivity on a latent-normal threshold scale. These null hypotheses are often similar, but they are not identical. Moreover, when the substantive claim concerns observed probabilities, accuracy, endorsement, or risk, the product-term coefficient is not itself the response-scale interaction. The response-scale interaction must be stated as a contrast, marginal effect, or difference between predicted probabilities (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019).

This fitted-data example is deliberately simple. Its purpose is not to show that logit and probit usually lead to different conclusions. They often do not. The point is more specific: small differences in link curvature can be hard to see in the fitted probability curves, yet remain structured rather than random. If a moderation claim depends on the part of the predictor range where this discrepancy is largest, especially near a floor, ceiling, chance level, or tail, link choice can become inferentially consequential.

This is why logit-versus-probit sensitivity should not be treated as a cosmetic robustness check. If a moderation claim appears only under one plausible link, changes sign under another, or depends on predictions near a boundary, the claim should be reported as conditional on the chosen link. Conversely, if the same response-scale conclusion is stable across logit, probit, and other theoretically defensible links, the interaction claim becomes easier to defend. The point is not that logit and probit usually disagree. The point is that an interaction claim is a claim about a no-interaction baseline, and the link defines that baseline.

The same logic applies beyond the logit-probit pair. A binomial family specifies a sampling model for successes out of trials, but it does not decide whether additivity is most plausible on a logit, probit, complementary log-log, flexible generalized-logistic, or chance-corrected scale. Classical work on binary-response links already treated alternative links as alternative scales for additivity, including symmetric and asymmetric departures from the logistic link (Aranda-Ordaz, 1981; Stukel, 1988). In forced-choice tasks, the link question includes whether the lower asymptote should be 0, as in a standard binomial link, or the theoretical chance level of the task. An ordinal family does not decide which cumulative link is appropriate, and applied ordinal-regression work explicitly treats logit, probit, complementary log-log, negative log-log, and cauchit links as different modeling choices (Smith et al., 2020). A Gamma family does not decide whether the mean should be additive on the identity, log, or inverse scale. For interaction testing, these choices are not merely alternative parameterizations. They define different scales on which no interaction is assumed.

**When link choice may matter less**

The link-function problem does not imply that every link choice will change the substantive conclusion. In many applications, plausible links lead to nearly identical fitted values and similar response-scale summaries. The risk is smaller when predicted probabilities remain far from floor and ceiling, effects are modest, the relevant predictor range is well observed, and candidate links produce almost overlapping curves in that range. As Figure 4 illustrates, the relevant issue is not whether two fitted curves look dramatically different. Often they do not. The issue is whether the pointwise discrepancy between them overlaps with the region that drives the interaction claim. Link choice is also less threatening when the interaction is reported as a response-scale quantity, such as a marginal effect or contrast, and when that quantity is robust across theoretically defensible links (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019). Some theories also do not depend on tail behavior. If the claim concerns average differences in a central range, rather than saturation, guessing, floors, or ceilings, link sensitivity may be small.

This is not a reason to ignore the link. It is a reason to report sensitivity analyses, so that readers can see whether the interaction claim is tied to a narrow modeling convention or survives plausible alternatives.

## Diagnostics Are Necessary but Not Sufficient

Diagnostics should be part of link-aware modeling, but they cannot settle the link-function question by themselves. A wrong link can generate a pseudo-interaction that becomes statistically detectable, while the diagnostics used to identify the wrong link may still have limited power or may flag only broad misfit. In that situation, the interaction test can become sensitive to the problem before the diagnostic evidence becomes specific enough to identify its source.

Pregibon-style link tests are useful because they target link adequacy more directly than generic residual plots (Pregibon, 1980). They expand the fitted GLM by adding terms derived from the fitted link, then ask whether the expanded model improves the fit. A significant added-term test suggests that the assumed link may be inadequate, but it does not identify the correct link. The signal may also reflect a missing nonlinear predictor, an omitted covariate, an omitted interaction, or another problem in the systematic component. A non-significant result is also limited. It means that this diagnostic, in this sample, did not detect this particular departure.

Simulation-based residual diagnostics provide a broader check. DHARMa implements randomized quantile residuals by simulating new responses from the fitted model and comparing the observed response with the fitted predictive distribution (Dunn & Smyth, 1996; Hartig, 2024). These residuals are easier to interpret across GLMs and GLMMs than raw, Pearson, or deviance residuals. In the diagnostic simulation below, we use four DHARMa checks: residual uniformity, dispersion, residual quantile patterns over fitted values, and residual quantile patterns over the focal predictor. These checks can detect broad misfit, dispersion problems, and structured residual patterns. But they do not by themselves distinguish a wrong link from a wrong family, a missing nonlinearity, a missing interaction, or a wrong measurement metric.

Model comparison tools, such as AIC, BIC, WAIC, or LOO, add another source of evidence, but only within a clear candidate set. For link diagnostics, their cleanest use is to compare candidate links while holding fixed the observed outcome, likelihood family or support, and structural formula. In that use, AIC can compare Poisson-log and Poisson-identity fits to the same counts, binomial-logit and binomial-probit fits to the same successes and trials, or Gamma-log and Gamma-inverse fits to the same positive outcome. It can also compare a standard binomial link with a chance-corrected binomial link when both define likelihoods for the same binomial data. It is less direct when the comparison changes the outcome, the likelihood support, or the measurement metric, which is why we do not treat sum-score alternatives as a simple AIC contest among links. More importantly, information criteria compare predictive fit among the models supplied by the analyst. They do not decide which scale defines the scientific no-interaction claim. A wrong or incomplete link may even be favored when the product term absorbs part of the curvature created by the misspecification. Thus, a lower AIC is evidence about relative fit in the candidate set, not proof of link adequacy.

For interaction claims, diagnostics should therefore be used as part of the argument, not as a final answer. They can show that a model is strained, but they cannot decide which scale is right for the scientific claim. If the claim concerns response-scale moderation, researchers should report response-scale contrasts or marginal effects and examine whether those quantities are stable across plausible links (McCabe et al., 2022; Mize, 2019; Rohrer & Arel-Bundock, 2026). If the claim concerns additivity on a latent or linear-predictor scale, the chosen link needs a substantive justification.

## Simulation Studies

The previous sections established the conceptual problem: the link function defines the scale on which a no-interaction model is additive. The simulations ask how this conceptual problem translates into inferential risk in settings where the true data-generating process is known. Across simulations, the true product-term interaction is set to zero on the generating scale. We then fit models that impose different scales of additivity and record how often they detect a statistically significant interaction. The target is not to estimate how often published interactions are false. It is to quantify when wrong or inadequate links can generate pseudo-interactions under controlled conditions (Domingue et al., 2024; Rimpler et al., 2026).

**Pseudo-interactions as power against the wrong null**

A link-induced false-positive interaction is not only a sampling accident. Under the correctly specified GLM link, the no-interaction null states that effects are additive on the scale defined by that link. Under a wrong link, the same expected means may imply a nonzero interaction on the fitted link scale. A significant product term can therefore be understood as power to detect a pseudo-interaction created by the mismatch between the generating link and the fitted link.

This idea extends the classic scale-dependence argument for interactions (Cox, 1984; Loftus, 1978; Wagenmakers et al., 2012) and connects it to work on GLM and outcome-model misspecification (Domingue et al., 2024). It also complements work showing that product-term coefficients in nonlinear GLMs do not automatically equal interaction effects on the response scale (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019).

In a simple 2 Ã— 2 design, let $\mu_{00}, \mu_{10}, \mu_{01}, \mu_{11}$ be the four expected response means generated with no product-term interaction on the target link scale. If the fitted model uses link $g$, the induced interaction on that fitted scale is

$$\Delta_g = g(\mu_{11}) - g(\mu_{10}) - g(\mu_{01}) + g(\mu_{00}).$$

When $g$ is the target link, $\Delta_g = 0$. When $g$ is not the target link, $\Delta_g$ can differ from zero even though the generating model contains no product term. The risk depends on the size of this pseudo-effect relative to its standard error. This is why link misspecification is often small in central ranges where plausible links are nearly equivalent, but larger near floors, ceilings, chance levels, or distributional tails.

The box summarizes the logic behind the simulation estimand. The reported detection rates are therefore interpreted as misspecification-induced detection rates, not as estimates of ordinary Type I error under a correctly specified GLM.

The simulations quantify two quantities. First, for each substantive scenario, we estimate the probability of detecting an interaction when the generating model contains no product term on the relevant scale. Second, in the diagnostic simulation, we estimate how often residual checks, a Pregibon-style link check, and same-formula AIC comparisons across candidate links flag the same misspecification that produced the pseudo-interaction. This comparison matters because link-oriented tests, simulation-based residual diagnostics, and information criteria can be useful, but they do not guarantee that the fitted link is adequate for the interaction claim being tested (Dunn & Smyth, 1996; Hartig, 2024; Pregibon, 1980).

The simulation code is intentionally separated from the manuscript. The main text reports only compact summaries. The full data-generating processes, scripts, and extended output are reported in the Supplement and repository.

**Simulation 1: Forced-Choice Accuracy**

Simulation 1 quantifies the forced-choice problem under controlled conditions. We generated binomial accuracy data with a .50 chance floor and no age-by-group product term on the chance-corrected logit scale. We then fit four models: a Gaussian identity model for proportions, standard binomial-logit and binomial-probit models, and a chance-corrected binomial-logit model. The outcome of interest was the false-positive rate for the age-by-group product term, that is, how often each fitted model detected an interaction that was absent on the generating scale.

*Note.* Successful fits are the number of replications in which the model converged and returned an interaction test. Estimates for models with failed fits are summarized over successful replications only.

The simulation is designed to be realistic rather than maximal. The sample size and number of trials are moderate, and the scenarios vary the overall performance level so that the wrong-link problem is visible without forcing all misspecified models to fail almost always. The expected pattern is that the chance-corrected model remains close to nominal alpha when it can be fitted, because it matches the scale used to generate the data.

The lower-performance scenario also shows a practical cost of using a link that encodes the task structure. The chance-corrected model is the target model in this simulation because the data were generated with a .50 lower asymptote, but it was harder to estimate when expected performance was close to that asymptote. This is a practical limitation of the proposed solution, not evidence that the standard binomial logit is preferable. In our summaries, results for this model should therefore be read as conditional on successful estimation. In applied analyses, researchers should report convergence failures, inspect whether failures concentrate near floors or ceilings, and use sensitivity analyses across plausible links. When the chance level is known from the task design, fixing it can help. Bayesian implementations in `brms` or Stan, with weakly informative priors, may also be useful as a sensitivity analysis because they can regularize unstable boundary behavior. Such models still require standard convergence checks and posterior predictive checks.

The Gaussian identity model and the standard binomial links can show elevated false-positive rates, especially when the scenario places one group closer to the chance floor. Standard binomial models do not always produce false interactions. When the theoretical floor is .50 but the fitted link treats 0 as the lower asymptote, however, the product term may become a way to repair a wrong no-interaction baseline.

**Simulation 2: Sum Scores**

Simulation 2 makes the measurement-scale problem concrete. We generated data on a latent continuous scale with a continuous predictor, a group effect, and no true x-by-group product term. We then converted the latent variable into ordinal item responses through item thresholds and summed the items. The interaction was therefore absent on the latent generating scale, but the observed score was bounded, discrete, and differentially compressed near the floor and ceiling.

The simulation compares three analyses, but only to separate the relevant scales. The observed sum-score identity model is the familiar Gaussian identity analysis of the manifest total. The Gaussian-logit bounded-score model treats the rescaled total as a bounded outcome and asks whether changing the scale of additivity changes the interaction conclusion. This model is not a recommended measurement model for sum scores. It is a crude sensitivity analysis for the manifest total, because the item responses are ordinal, bounded, and heterogeneous. The latent generating scale is an idealized benchmark available only in the simulation, because the true latent theta is known. It is included to show what the interaction test would target if the generating construct scale were directly observed.

The expected pattern is not that sum scores always create false interactions. The point is conditional. When the manifest score is approximately linear over the observed range, far from its bounds, and the claim is about raw-score units, the Gaussian identity analysis may be a reasonable approximation. When the score compresses the latent scale unevenly, the raw-score no-interaction baseline can diverge from the latent no-interaction baseline. In that case, a statistically significant product term can reflect the score metric as much as, or more than, moderation of the underlying construct. When the claim concerns an interaction between latent constructs, the more direct solution is not to search for another link for the manifest total, but to model the measurement structure explicitly, for example with SEM, item-level ordinal models, IRT-inspired models, or other explicit measurement models. These models still require assumptions about thresholds, links, loadings, and latent-response metrics, so they do not remove the scale problem. They make the measurement part of the interaction claim explicit. The practical conclusion is to treat sum-score analyses as sensitivity analyses for manifest-score claims, not as substitutes for latent-variable modeling.

### Simulation 3: Within-Family Link Choice

Simulation 3 moves from the fitted-data illustration in Figure 4 to a repeated-sampling question: can within-family link choice inflate false-positive interaction rates when the outcome family, experimental design, and random-effect structure are held fixed? We generated repeated binary responses in a two-group, two-condition design, with 15 trials per participant-condition cell and a subject random intercept. The true condition-by-group product term was zero on the generating link scale. We then crossed the generating link and fitted link, using logit and probit in both roles, and fitted all models as random-intercept GLMMs.

This simulation is deliberately narrow. It does not compare a correct binary model with a Gaussian identity model, and it does not change the random-effects structure. The only misspecification is within-family link choice. The target null is no condition-by-group product term on the generating link scale. Under the wrong fitted link, the same cell probabilities can imply a small nonzero product term on the fitted link scale. With enough information, that deterministic mismatch can become statistically detectable.

The point is not that logit and probit usually lead to different conclusions. They often do not, especially when probabilities are concentrated in the middle of the response range. The point is more specific: when the design places one group or condition closer to a tail, floor, or ceiling, a plausible but wrong link can turn a no-interaction data-generating process on one scale into an apparent interaction on another. Simulation 3 quantifies this risk, whereas Figure 4 shows why the visual similarity of fitted curves is not, by itself, a guarantee of interaction robustness.

The table reports false-positive rates relative to the generating-link null. The median product-term coefficient and its standard error are on the fitted link scale, so they should not be read as response-scale changes in probability. The figure makes the Monte Carlo result explicit: even with the same binary outcome family, logit and probit define slightly different no-interaction baselines, and this difference can become inferentially visible in repeated samples. A result that is stable across both links is easier to defend. A result that appears only under one link should be reported as link-conditional, or translated into a response-scale estimand and checked across plausible links.

**Diagnostic Simulation**

Finally, we examined whether common diagnostics would flag the same link problems that produced pseudo-interactions. This simulation is not meant to be a general benchmark of all possible checks. It is tied to the paper's main question: when the fitted link defines the wrong scale of additivity, does the diagnostic evidence become visible before, after, or at the same time as the false interaction?

We used five scenarios, each tied to a previous example or matched counterpart in the paper. The first is the simple count example in Figure 3. Data are generated from a Poisson-log model in which expected errors decline with age, with no age-by-group product term on the log-count scale. The coefficients are the same as in the figure: intercept 2.60, age effect -0.55 per year after age 6, and group effect 0.65. The fitted model keeps the Poisson family and the interaction formula, but uses an identity link. This makes the comparison a same-family link comparison, not a comparison between Poisson and Gaussian models.

The second scenario is the lower-performance forced-choice scenario from Simulation 1. Data are generated with a .50 chance floor, 20 trials per participant, intercept -0.80, age effect 0.60, group effect -0.90, and no age-by-group product term on the chance-corrected logit scale. The fitted model is a standard binomial-logit interaction model, which respects the binomial trial structure but places the lower asymptote at 0 rather than at chance.

The third scenario mirrors the probit-generated, logit-fitted case in Simulation 3. Data are repeated binary trials with a subject random intercept, 15 trials per condition cell, and latent ICC .30. The fixed effects are the probit reference values used in Simulation 3: intercept 1.50, group effect -1.00, condition effect -1.00, and no group-by-condition product term. The fitted model is the corresponding random-intercept logit GLMM with the same interaction formula.

The fourth scenario is a continuous-predictor counterpart to the previous binary repeated-trials case. It keeps the probit data-generating link, the logit fitted link, the group effect, the focal-predictor effect, the subject random intercept, and the latent ICC, but replaces the two-level condition with a continuous predictor sampled from 0 to 1. This scenario asks whether the same logit-versus-probit issue is visible when the focal predictor has enough unique values for continuous-predictor residual checks to be better identified.

The fifth scenario connects to the Gamma example in Figure 1. Data are generated from a Gamma model with a log link, shape 8, baseline mean 400 when x = 0 and group = 0, x effect 0.25 on the log-mean scale, group effect 0.25, and no x-by-group product term. The fitted model keeps the Gamma family and the interaction formula, but uses the inverse link.

In each replication, diagnostics were applied to the misspecified interaction model, because this is the model an analyst would usually inspect after finding the interaction. This choice makes the diagnostic test more realistic and also more demanding. If the wrong link has induced a product term and the product term has absorbed part of the wrong-link pattern, residual and link checks may have less structure left to detect. The simulation therefore asks whether the same workflow that produces the interaction would also warn the analyst that the scale is questionable.

The residual workflow used four DHARMa checks: residual uniformity, dispersion, residual quantile patterns over fitted values, and residual quantile patterns over the focal predictor (Dunn & Smyth, 1996; Hartig, 2024). We also used a Pregibon-style added-term link check as a more link-oriented diagnostic (Pregibon, 1980). Finally, we used AIC only as a same-formula link comparison. In each scenario, the target-link interaction model was compared with the fitted-link interaction model. Thus, AIC varied the link, not the presence or absence of the product term. When both AIC values were available, the model with lower AIC was counted as the AIC winner. We did not use an inconclusive category, so these columns should be read as strict lower-AIC counts, not as strong evidence thresholds.

We keep the figure for the chance-floor scenario because it is the most distinctive psychological case. It also shows why a diagnostic table alone is not enough: the generating pattern is simple and additive on the chance-corrected link scale, but the standard binomial link forces the curve toward a lower asymptote of 0 rather than .50.

These results are the warning rather than a contradiction. Diagnostics can help, and same-formula AIC comparisons can be informative when the candidate links are substantively plausible. But they do not decide the scale of additivity. Residual checks may have limited power, may flag broad misfit without identifying its source, and may become weaker after a pseudo-interaction has absorbed part of the wrong-link pattern. AIC is similarly relative: it chooses among the candidate models supplied by the analyst, and the lower-AIC model may still be a poor representation of the scientific no-interaction baseline. For interaction claims, diagnostics should therefore be paired with a clear estimand, a justified link, and sensitivity analyses over plausible links.

## A Minimal Workflow for Link-Aware Interaction Testing

The practical implication is not that researchers should stop testing interactions. It is that interaction claims should be made scale-aware. A minimal workflow is shown in Table 8.

The workflow is not a rule that makes interaction testing automatic. It is a compact reporting standard. Researchers still need theory to decide which contrast matters, which scale is meaningful, which links are plausible, and whether the design is informative enough for the interaction claim (Baranger et al., 2023; Castillo et al., 2026). The table asks that these decisions be visible before the product term is interpreted. A linear-predictor coefficient may be useful for a model-based claim, whereas a response-scale contrast or marginal effect may be closer to the substantive claim made in words (McCabe et al., 2022; Mize, 2019; Rohrer & Arel-Bundock, 2026). Diagnostics, predictive checks, and information criteria can warn about misspecification or relative fit, but they should be treated as secondary checks rather than as validation of the interaction claim (Dunn & Smyth, 1996; Hartig, 2024; Pregibon, 1980). The central robustness question is whether the target estimand remains substantively stable across a small, theory-limited set of plausible links or measurement models. This is related to the logic of multiverse analysis, but the recommendation here is narrower: a targeted sensitivity analysis defined by the outcome, the task, and the scientific estimand. When the interaction conclusion is stable across plausible specifications, the claim becomes easier to defend. When it is not stable, the conclusion should be stated as conditional on a modeling choice. This extends a basic reporting principle: choices that affect the claim should be visible, justified, and interpretable (Mustillo et al., 2018).

For example, a treatment for panic attacks may reduce the expected count by the same proportion in mild and severe patients. On a log-count scale, this is consistent with no product-term interaction. On the observed count scale, severe patients may still show a larger absolute reduction because they start with more attacks, whereas mild patients may already be close to zero. The response-scale contrast is useful for questions about patient benefit and resource allocation. The link-scale result is useful for claims about the form of the modelled process. The two quantities answer different questions.

## Discussion

Interactions remain central to psychological theory, and the point of this paper is not that they are usually wrong. Some interaction claims will be stable across reasonable links, scales, and response-scale summaries. Others will not. Our claim is narrower: many interaction claims are incomplete unless they state the scale on which additivity, and therefore no interaction, is being assumed. This scale problem is well known (Cox, 1984; Loftus, 1978; Wagenmakers et al., 2012), but in GLMs it has a practical form: the link function defines the scale on which the model is additive. The same issue may also extend beyond interactions to nonlinear effects, because many psychological analyses search for curvature, thresholds, and changing slopes without first specifying the scale on which these patterns are expected (Domingue et al., 2024; McCabe et al., 2022; Rohrer & Arslan, 2021).

The preregistered review showed why this matters for current practice. Interaction tests were common, but link functions were often implicit, and many interaction-testing articles included potentially questionable identity-link analyses, that is, identity-link analyses with outcomes for which observed-scale additivity was not self-evident. These findings do not show that the reviewed articles reached false conclusions. They show that the scale of many interaction claims is difficult to evaluate from the published report.

The conceptual framework separated three scale-related problems, while centering the second one. A wrong-family problem arises when the model does not respect the outcome support or sampling process. A wrong-link problem arises when the family is broadly plausible, but the link defines an inadequate scale of additivity. A wrong-metric problem arises when the recorded score is not the scale on which the theoretical interaction claim is meant to hold. The main focus of this paper is the wrong-link problem. Forced-choice accuracy illustrates it clearly: a standard binomial family respects trial counts, but a standard link may fail to encode the non-zero chance floor of the task. Within-family links illustrate the same point more generally: even plausible links can define different baselines for no interaction. Sum scores are included as a related measurement case, because manifest scores can introduce thresholds, bounds, and compression that change interaction claims. For this reason, the sum-score example should be read as a measurement-metric warning, not as a recommendation to replace item-level or latent-variable models with a transformed total score. In this sense, ordinary floor and ceiling effects are not a separate problem. They are the visible extreme of a broader remapping problem.

The simulations were designed to quantify pseudo-interactions, not to estimate the prevalence of false interactions in the literature. They should therefore be read alongside, rather than as a replacement for, work showing that interaction estimates are often difficult to estimate precisely even when the model form is appropriate (Baranger et al., 2023; Castillo et al., 2026). They show how link and metric choices can generate significant product terms even when the true interaction is zero on the generating scale. The diagnostic simulation adds a practical caution. Residual checks, Pregibon-style tests, and information criteria can flag some forms of misspecification, and they should be used. However, they rarely answer one clean question. They may miss the problem, detect only broad misfit, or prefer the better-fitting model among an incomplete candidate set. They should therefore not be read as proof that an interaction claim is robust to link choice (Dunn & Smyth, 1996; Hartig, 2024; Pregibon, 1980).

Several limitations are important. First, the review was descriptive and targeted to selected journals, not a census of psychological research. Second, we did not code whether published interactions were removable or nonremovable, because that distinction often cannot be recovered from published reports alone. Third, the simulations are intentionally simple. More realistic data may include random effects, item heterogeneity, lapses, response bias, nonlinear predictors, and missing data. Fourth, chance-corrected links are appropriate only when a chance floor is theoretically justified. They are not a universal replacement for standard binomial models. In forced-choice tasks, the chance floor may be fixed by design, estimated under stronger assumptions, or varied in sensitivity analyses. Finally, logit and probit links often lead to very similar conclusions, especially away from boundaries; our claim is that link choice can matter for interaction claims, not that it always does.

The main contribution is therefore practical. Link-aware interaction testing does not make moderation claims automatic, and it does not identify one universally correct scale. It asks researchers to make the main judgments visible: what outcome family is plausible, what link defines the no-interaction baseline, what response-scale quantity answers the substantive question, and whether the conclusion survives plausible alternatives. This workflow complements existing guidance on nonlinear interactions, marginal effects, and prediction-based interpretation (McCabe et al., 2022; Mize, 2019; Rohrer & Arel-Bundock, 2026). When the same conclusion holds across defensible links, the interaction claim is strengthened. When it changes, the result can still be informative, but it should be reported as conditional on a modeling choice. In GLMs, choosing the link is choosing the scale of the no-interaction claim.

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**Table 1**

*Descriptive summary of the preregistered review. Potentially questionable identity-link analyses are cases where observed-scale additivity was not self-evident from the reported outcome.*

| Quantity | n | Denominator | Percent | 95% CI |
|---|---|---|---|---|
| Eligible empirical articles reviewed | 331 | 331 | 100.0% | [98.9%, 100.0%] |
| Articles testing at least one interaction | 200 | 331 | 60.4% | [55.1%, 65.5%] |
| Interaction-testing articles using a non-identity link | 58 | 200 | 29.0% | [23.2%, 35.6%] |
| Interaction-testing articles explicitly reporting the link function | 46 | 200 | 23.0% | [17.7%, 29.3%] |
| Interaction-testing articles with potentially questionable identity-link analyses | 144 | 200 | 72.0% | [65.4%, 77.8%] |
| Potentially questionable identity-link cases with at least one significant interaction | 124 | 144 | 86.1% | [79.5%, 90.8%] |

**Table 2**

*Broad response-variable categories in interaction-testing articles. Categories are non-exclusive, so one article can contribute to more than one row.*

| Outcome type | n | Denominator | Percent |
|---|---|---|---|
| Binary / accuracy / proportions | 86 | 200 | 43.0% |
| Sum scores / composites | 77 | 200 | 38.5% |
| Ordinal / Likert / ratings | 49 | 200 | 24.5% |
| Response times / durations | 29 | 200 | 14.5% |
| Neural / physiological measures | 23 | 200 | 11.5% |
| Difference / distance scores | 21 | 200 | 10.5% |
| Counts / error counts | 12 | 200 | 6.0% |
| Correlations / associations | 8 | 200 | 4.0% |

**Table 3**

*Three sources of scale-related problems for interaction claims.*

| Problem | Key question | Psychological example | Risk for interaction claims |
|---|---|---|---|
| Wrong family | Does the model respect the support and sampling process of the outcome? | Counts, proportions, or trial accuracies modeled as unbounded Gaussian observations. | Constraints, discreteness, or the mean-variance relation may be absorbed by a product term. |
| Correct family, wrong link | Given a plausible family, which scale defines no interaction? | Forced-choice accuracy modeled with a standard binomial logit, although the task has a known chance floor such as .50. | The model may test additivity on the wrong scale, so a pseudo-interaction can arise even when the target link-scale model has only main effects. |
| Wrong measurement metric | Is the recorded score an interval-scale measure of the target construct? | Likert items or sum scores treated as raw interval outcomes without justification. | A product term may reflect thresholds, bounds, or scale compression rather than moderation of the underlying construct. |

**Table 4**

*Simulation 1: false-positive interaction rates and median model-implied observed-scale changes in the group gap for forced-choice accuracy data. CI = Wilson 95% confidence interval.*

| Scenario | Model | Fits | Sig | False-positive rate | 95% CI | Median gap change |
|---|---|---|---|---|---|---|
| Lower performance | Gaussian identity | 300 | 137 | 0.457 | [0.401, 0.513] | -1.779 |
| Lower performance | Standard binomial logit | 300 | 191 | 0.637 | [0.581, 0.689] | -1.769 |
| Lower performance | Standard binomial probit | 300 | 183 | 0.610 | [0.554, 0.663] | -1.781 |
| Lower performance | Chance-corrected binomial logit | 178 | 14 | 0.079 | [0.047, 0.128] | -1.225 |
| Middle performance | Gaussian identity | 300 | 40 | 0.133 | [0.099, 0.176] | -0.865 |
| Middle performance | Standard binomial logit | 300 | 169 | 0.563 | [0.507, 0.618] | -0.862 |
| Middle performance | Standard binomial probit | 300 | 148 | 0.493 | [0.437, 0.550] | -0.866 |
| Middle performance | Chance-corrected binomial logit | 300 | 18 | 0.060 | [0.038, 0.093] | -0.707 |
| Higher performance | Gaussian identity | 300 | 36 | 0.120 | [0.088, 0.162] | 0.654 |
| Higher performance | Standard binomial logit | 300 | 103 | 0.343 | [0.292, 0.399] | 0.555 |
| Higher performance | Standard binomial probit | 300 | 59 | 0.197 | [0.156, 0.245] | 0.617 |
| Higher performance | Chance-corrected binomial logit | 300 | 14 | 0.047 | [0.028, 0.077] | 0.526 |

**Table 5**

*Simulation 2: sum scores as bounded and discrete outcomes. Panel A shows how a latent additive data-generating model, with no x-by-group product term, is mapped onto the observed sum-score scale under middle, floor, and ceiling scenarios. Panel B shows the response-scale group gap implied by the same latent group difference; the gap can change across x because the observed score is compressed unevenly. Panel C shows false-positive interaction rates across repeated simulations. The latent generating scale is an idealized benchmark available only because theta is known in the simulation, not a model available in ordinary applied data analysis.*

| Scenario | Model | Scale | Fits | Sig | False-positive rate | 95% CI | Median gap change |
|---|---|---|---|---|---|---|---|
| Middle of scale | Gaussian-logit bounded score | sum score | 300 | 13 | 0.043 | [0.025, 0.073] | -0.625 |
| Middle of scale | Latent generating scale | latent theta | 300 | 16 | 0.053 | [0.033, 0.085] | 0.018 |
| Middle of scale | Observed sum-score identity | sum score | 300 | 29 | 0.097 | [0.068, 0.135] | -0.547 |
| Near ceiling | Gaussian-logit bounded score | sum score | 300 | 15 | 0.050 | [0.031, 0.081] | 1.530 |
| Near ceiling | Latent generating scale | latent theta | 300 | 16 | 0.053 | [0.033, 0.085] | 0.003 |
| Near ceiling | Observed sum-score identity | sum score | 300 | 112 | 0.373 | [0.321, 0.429] | 1.328 |
| Near floor | Gaussian-logit bounded score | sum score | 300 | 23 | 0.077 | [0.052, 0.112] | -1.916 |
| Near floor | Latent generating scale | latent theta | 300 | 18 | 0.060 | [0.038, 0.093] | 0.011 |
| Near floor | Observed sum-score identity | sum score | 300 | 216 | 0.720 | [0.667, 0.768] | -1.762 |

**Table 6**

*Simulation 3: false-positive interaction rates for the condition-by-group product term in repeated-trial random-intercept GLMMs. CI = Wilson 95% confidence interval.*

| Generating link | Fitted link | Link status | Fits | Sig | False-positive rate | 95% CI | Median beta interaction | Median SE |
|---|---|---|---|---|---|---|---|---|
| logit | logit | Matched link | 300 | 16 | 0.053 | [0.033, 0.085] | -0.001 | 0.075 |
| logit | probit | Wrong link | 300 | 45 | 0.150 | [0.114, 0.195] | -0.041 | 0.042 |
| probit | logit | Wrong link | 300 | 41 | 0.137 | [0.102, 0.180] | 0.071 | 0.076 |
| probit | probit | Matched link | 300 | 15 | 0.050 | [0.031, 0.081] | -0.003 | 0.043 |

**Table 7**

*Diagnostic simulation across five link-misspecification scenarios. FP interaction reports how often the misspecified interaction model detected a product term absent on the generating link scale. DHARMa gives the range across prespecified checks, not an omnibus diagnostic. Pregibon reports the added-term link check. AIC compares the true-link and fitted-link interaction models while keeping the structural formula fixed; because no inconclusive Î"AIC category is used, AIC columns report only which candidate had the lower AIC. CI = Wilson 95% confidence interval.*

| Case | True link | Fitted link | FP interaction | DHARMa | Pregibon | AIC true | AIC fitted |
|---|---|---|---|---|---|---|---|
| Count errors | log | identity | 0.942 [0.909, 0.964] | 0.007-0.906 (4) | 0.996 | 0.996 | 0.004 |
| Chance-floor accuracy | chance-corrected logit | standard logit | 0.580 [0.523, 0.634] | 0.000-0.070 (4) | 0.277 | 0.000 | 1.000 |
| Binary repeated trials | probit | logit | 0.100 [0.071, 0.139] | 0.000-0.010 (3) | 1.000 | 0.930 | 0.070 |
| Binary continuous predictor | probit | logit | 0.073 [0.049, 0.109] | 0.000-0.020 (4) | 1.000 | 0.967 | 0.033 |
| Gamma mean response time | log | inverse | 0.280 [0.232, 0.333] | 0.000-0.080 (4) | 0.353 | 0.810 | 0.190 |

**Table 8**

*A minimal workflow for link-aware interaction testing.*

| Step | Key question | Minimal reporting |
|---|---|---|
| Define the estimand | What quantity answers the moderation claim? | State the contrast, moderator values, scale, and whether the estimand is descriptive, associational, or causal. |
| State the outcome model | What family matches the support and sampling process? | Report the outcome range, discreteness, trial or item structure, chance level, and family. |
| Justify the link | On which scale should effects add if there is no interaction? | Name the link and give the substantive or design reason, such as odds, latent threshold, log mean, or chance floor. |
| Report the interaction on the relevant scale | Is the claim about the link scale, the observed scale, or both? | Report predicted means, contrasts, marginal effects, or difference-in-differences, not only the product coefficient. |
| Probe robustness | Does the conclusion survive plausible alternatives? | Report the target estimand across plausible links or measurement models; report diagnostics as secondary checks, not as proof of link adequacy. |

**Figure 1**

*Same outcome family, different link functions. The grey construction lines are drawn from a target link: the chance-corrected logit in Panel A and the log link in Panel B. Vertical grey lines mark equal intervals on the target-link linear predictor scale. Horizontal grey lines show where those equally spaced target-link values fall on the observed response scale. The key point is that equal intervals on one link scale do not remain equal intervals under another link. Thus, models with the same outcome family can imply different no-interaction baselines, even when their fitted curves look similar over the observed range.*

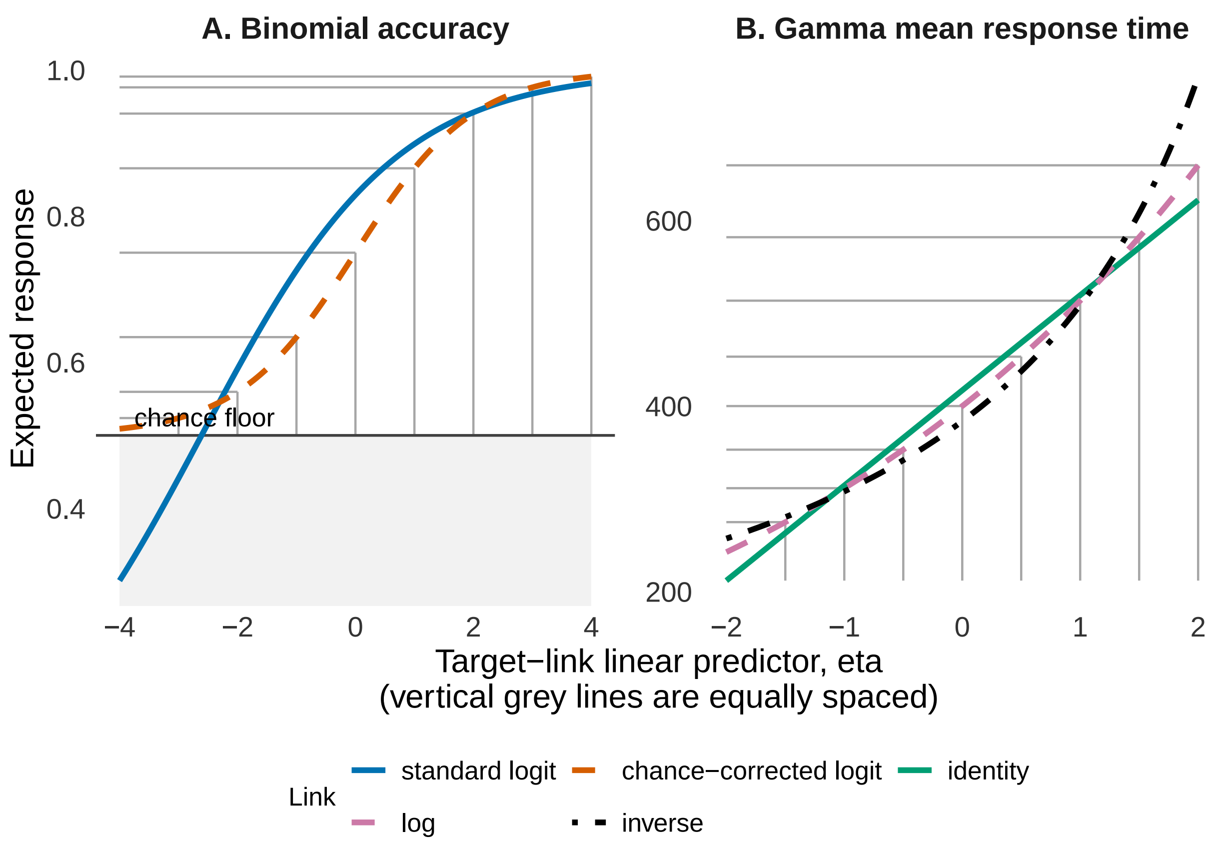

**Figure 2**

*Same data, different link functions. Data were simulated from a 2-alternative forced-choice accuracy task with 50 trials per participant and a 0.50 chance floor. The data-generating model included age and group effects, but no age-by-group interaction on the chance-corrected 2-AFC logit scale. Panel A shows the true response-scale pattern implied by that model. Panel B shows one simulated dataset. Panel C fits the same data with three models. The Gaussian identity model is simple and interpretable on the observed accuracy scale, but it can produce predictions outside the 0-to-1 range and ignores the binomial sampling process. The standard binomial logit/probit model, shown here with a logit link, respects the binomial process and the 0-to-1 range, but assumes a lower asymptote at 0. The chance-corrected binomial link incorporates the task's chance floor. The question is which scale defines the scientific null hypothesis of no moderation.*

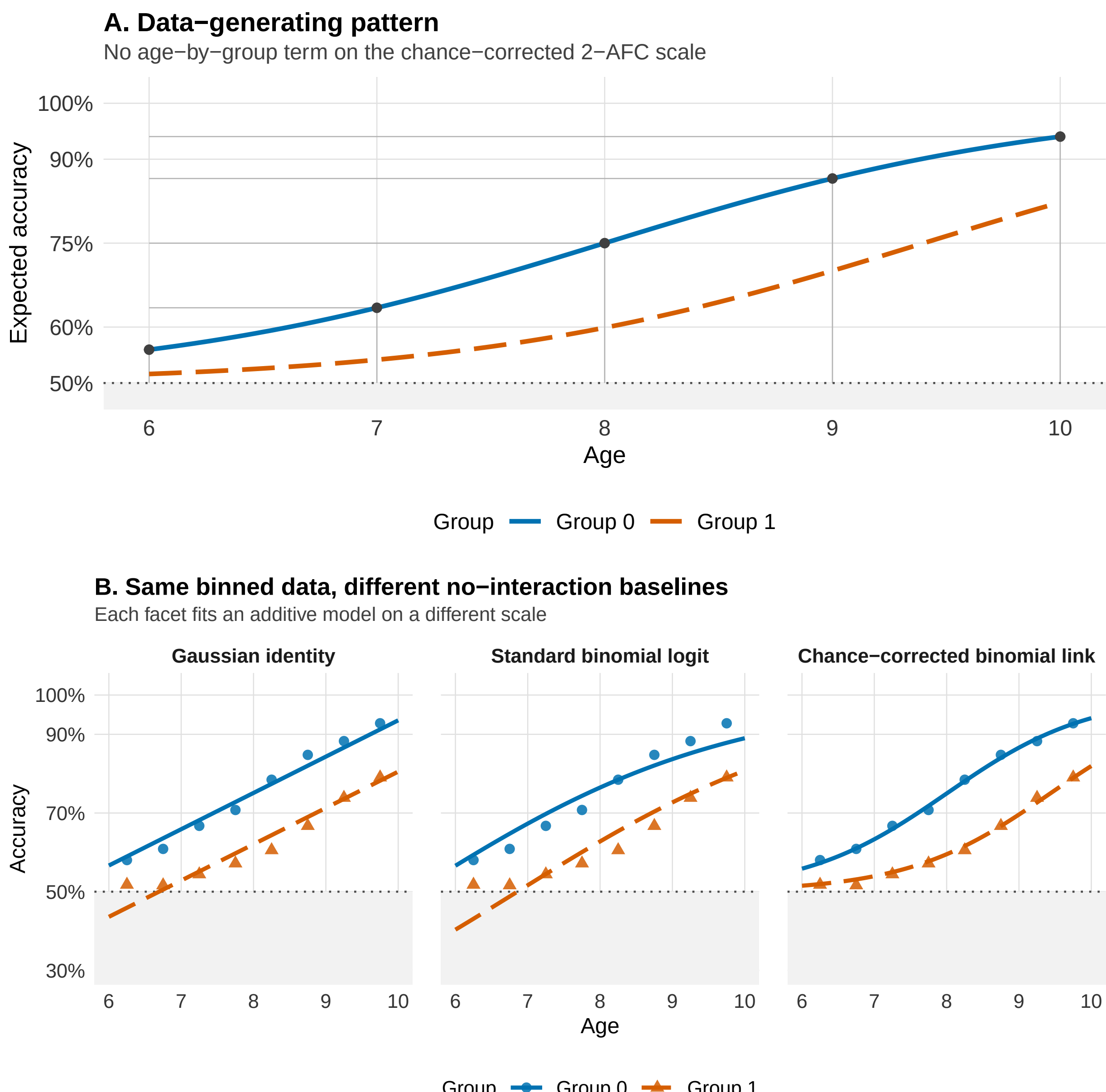

**Figure 3**

*A simple count example. The two groups/conditions have the same age effect on the log-count scale and no age-by-condition product term on that scale, but after exponentiating the linear predictor, expected error counts decrease nonlinearly on the observed count scale, and differences between conditions appear to change with age.*

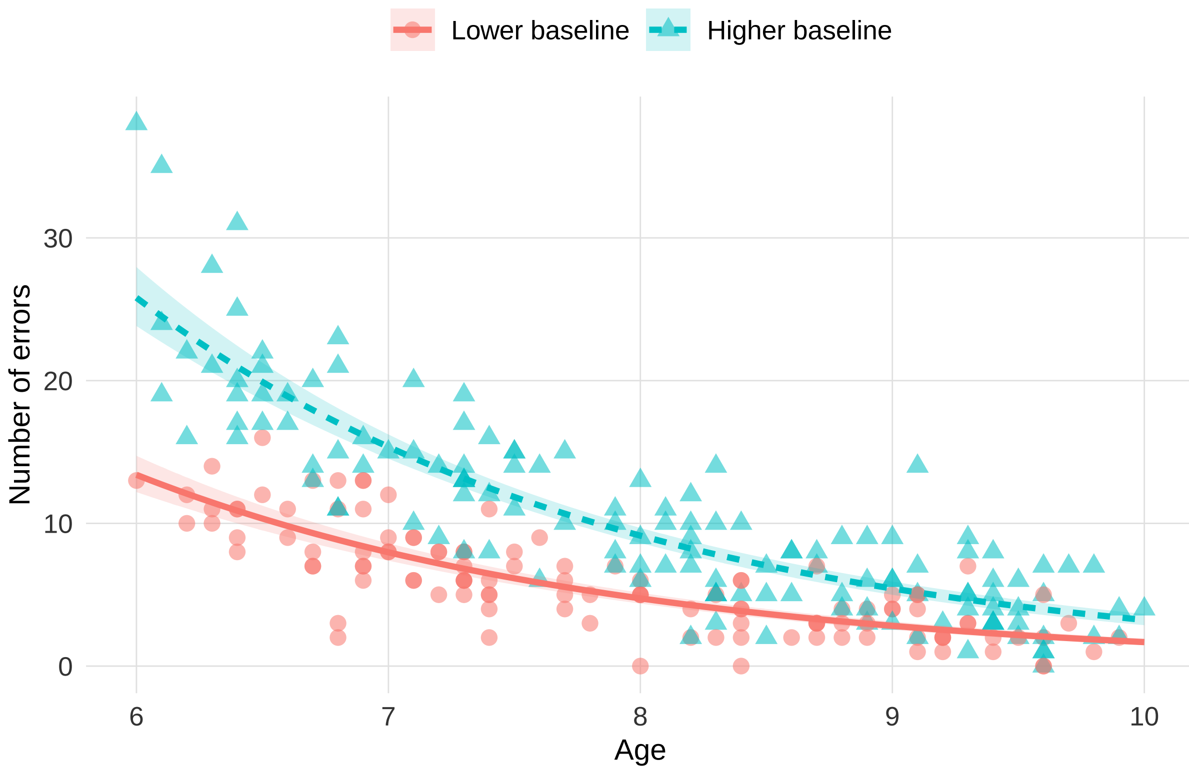

**Figure 4**

*Fitted logit and probit models on one simulated binary dataset. The data were generated from a logistic curve, and both a binomial-logit and a binomial-probit GLM were fitted to the same responses. Panel A shows binned observed proportions together with the two fitted probability curves. Panel B shows the pointwise difference between the fitted probabilities, computed as logit minus probit. The curves can look nearly interchangeable while still implying a structured discrepancy across the predictor range. For interaction claims, this matters when the relevant comparison depends on regions where the two fitted links allocate different response-scale room for an effect.*

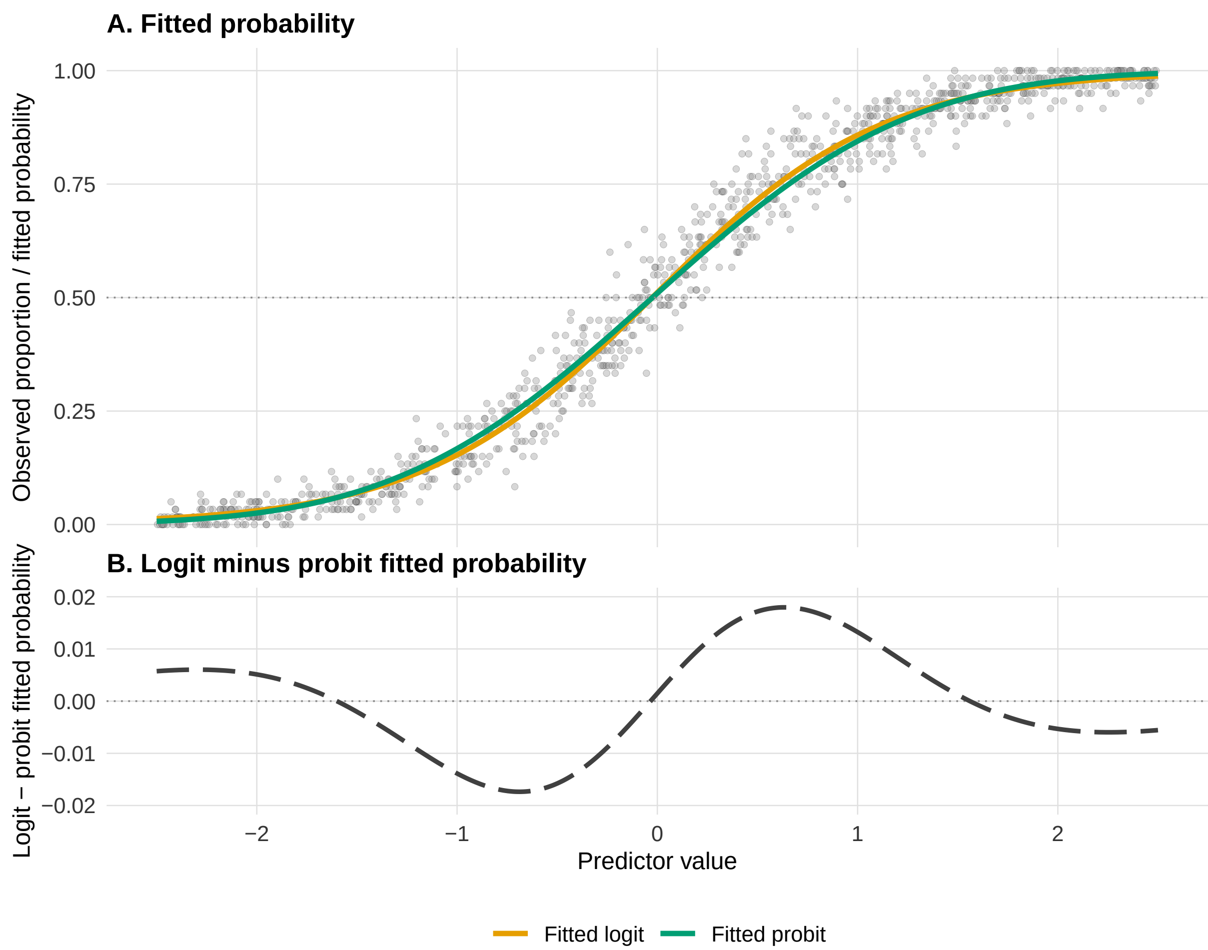

**Figure 5**

*Simulation 1: forced-choice accuracy with a task-implied chance floor. Panel A shows the deterministic scenario curves generated by the chance-corrected logit model. The grey region marks values below the .50 chance floor, and the dashed horizontal line marks chance performance. Panel B shows the false-positive rate for the age-by-group product term across repeated simulations. Data are generated with no age-by-group interaction on the chance-corrected scale.*

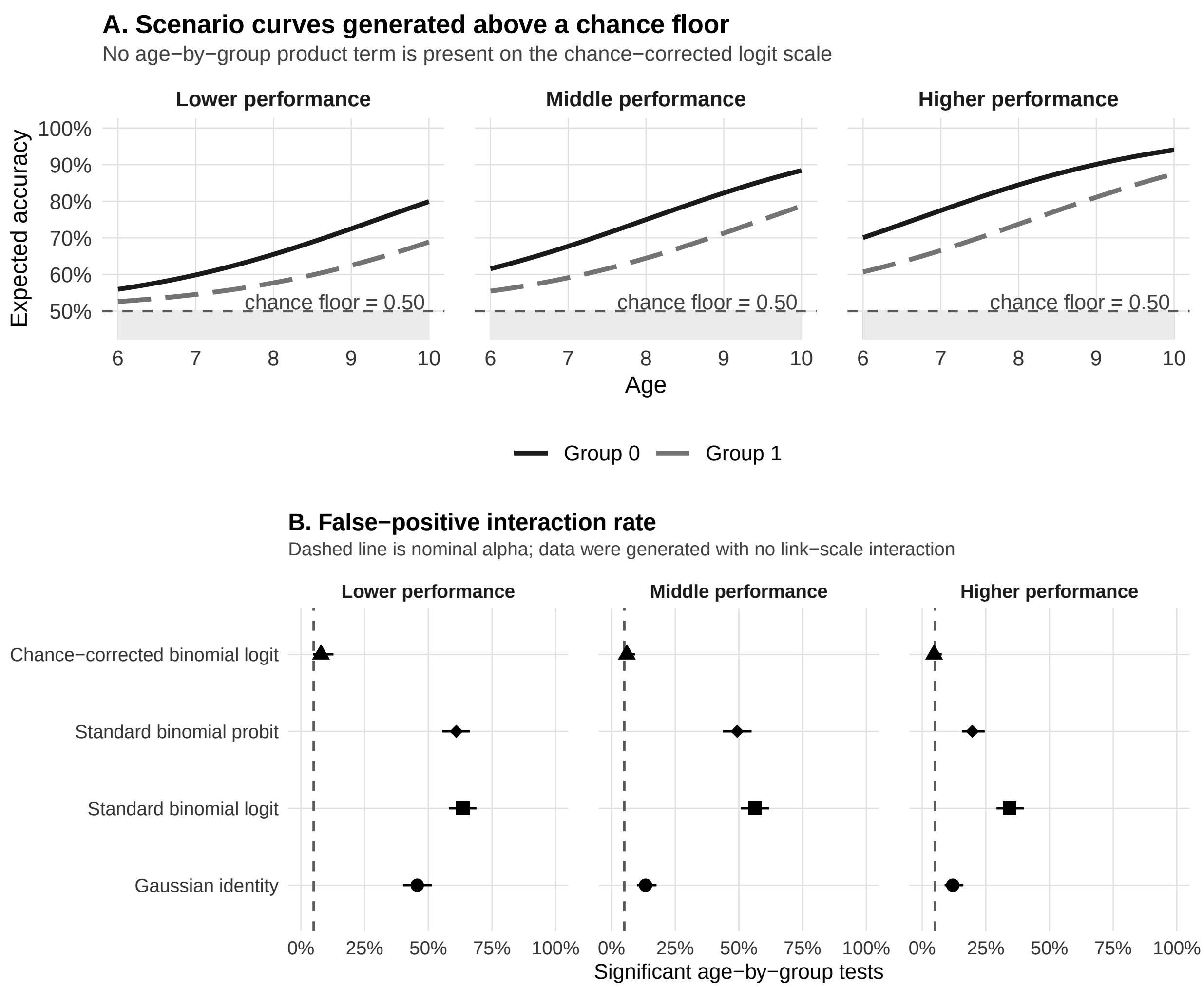

**Figure 6**

*Simulation 2: sum scores as bounded and discrete outcomes. Panel A shows how a latent additive data-generating model, with no x-by-group product term, is mapped onto the observed sum-score scale under middle, floor, and ceiling scenarios. Panel B shows the response-scale group gap implied by the same latent group difference; the gap can change across x because the observed score is compressed unevenly. Panel C shows false-positive interaction rates across repeated simulations. The latent generating scale is an idealized benchmark available only because theta is known in the simulation, not a model available in ordinary applied data analysis.*

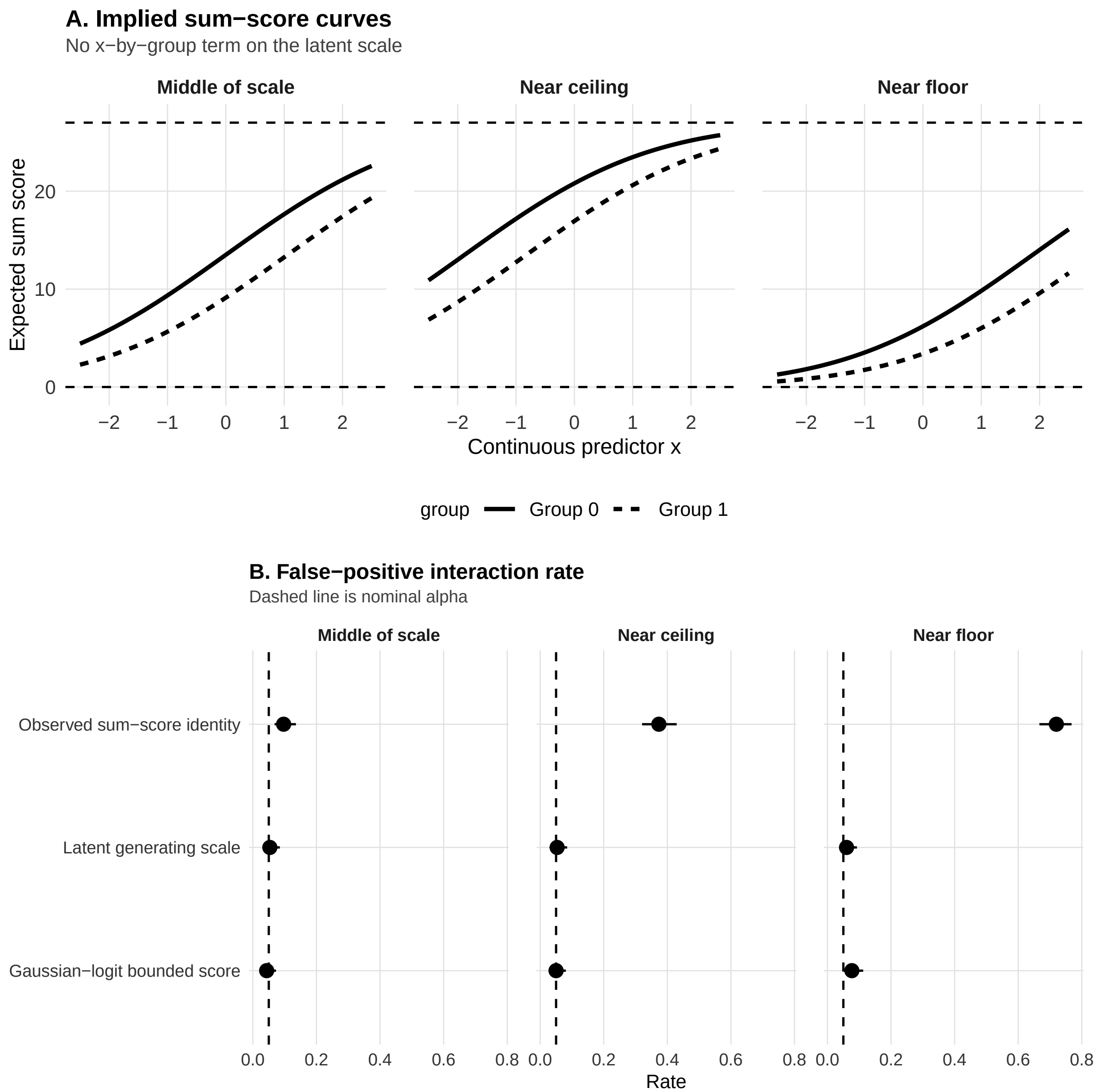

**Figure 7**

*Simulation 3: within-family link choice in repeated-trial random-intercept GLMMs. Panel A shows the deterministic cell probabilities generated by additive logit or probit models at random intercept 0. Panel B shows the false-positive rate for the condition-by-group product term when the generating and fitted links are crossed. Error bars are Wilson 95% confidence intervals, and the dashed line marks nominal alpha.*

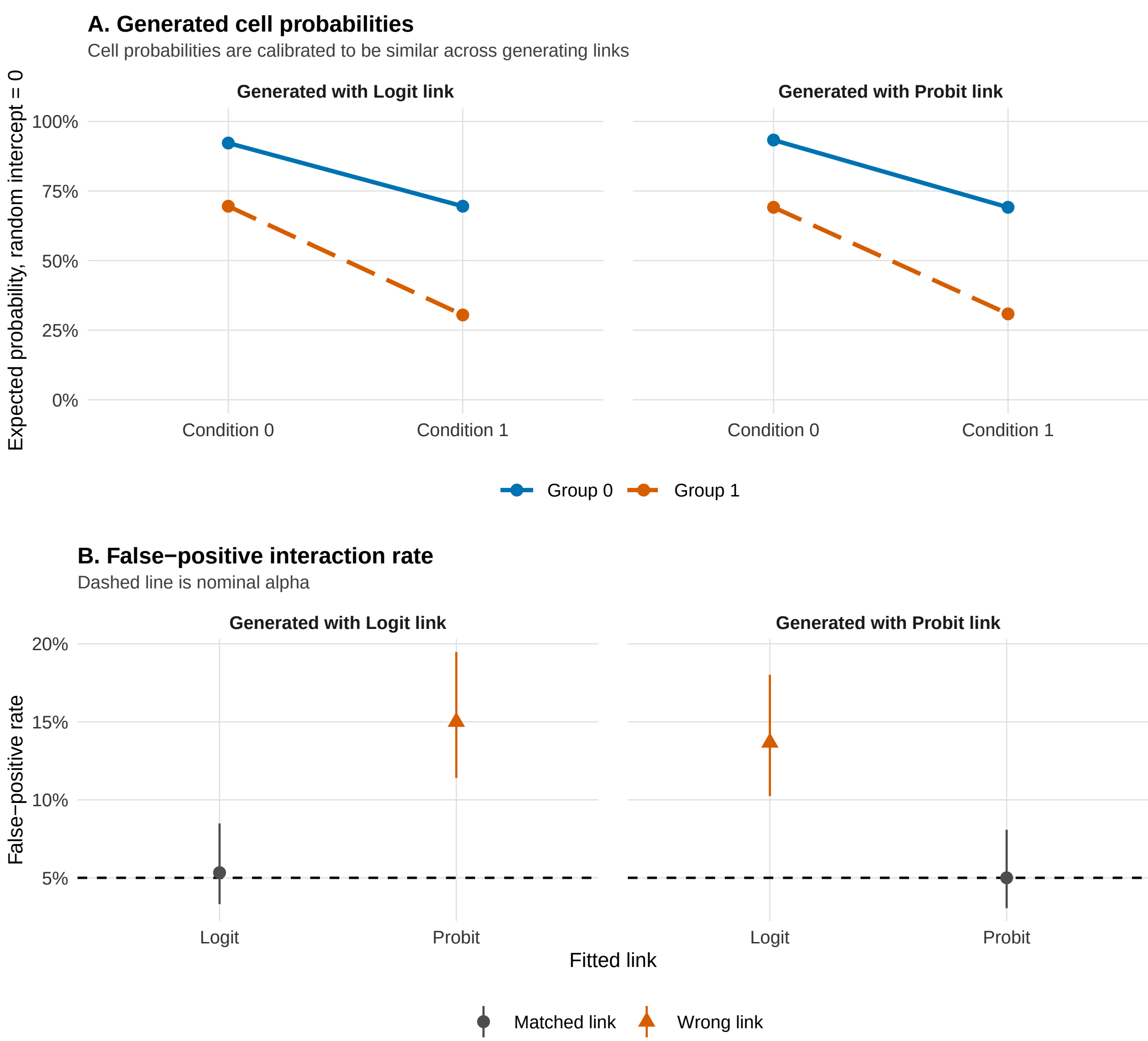

**Figure 8**

*Diagnostic simulation for the chance-floor scenario. Panel A shows the deterministic generating pattern for the lower-performance forced-choice scenario from Simulation 1: a chance-corrected binomial model with a .50 floor and no age-by-group product term on the generating link scale. Panel B shows the false-positive rate for the age-by-group interaction after fitting the wrong standard binomial-logit link. Panel C shows diagnostic detection rates reported separately, not as an omnibus model-validity test: DHARMa uniformity, DHARMa dispersion, DHARMa residual quantiles over fitted values, DHARMa residual quantiles over the focal predictor, and a Pregibon-style added-term link check. Error bars are Wilson 95% confidence intervals. Dashed lines mark nominal alpha for the tests; rates near alpha mean that the diagnostic did not often flag the wrong-link pattern in this scenario.*